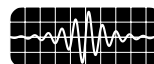


Nicolas Steven Miller

Software Architect, Principal Electrical Engineer

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SUMMARY

Lead architect of software ecosystem—compiler, simulator, firmware, Python SDK, ML pipeline—for state-of-the-art field-programmable analog array (FPAA) chip at a semiconductor startup. 15 years shipping production systems at startups spanning software, embedded hardware, and semiconductors.

SKILLS

- **Languages:** Rust, Python, C, C++
- **Embedded Systems:** PCB Design, Microcontrollers, STM32, Firmware, RTOS, Serial Protocols
- **Electrical Engineering:** Signal Processing, Analog Circuit Design, Mixed-Signal Integrated Circuits, Semiconductor ATE, Test Equipment (Oscilloscope, Logic Analyzer, Waveform Generator)
- **Machine Learning & AI:** PyTorch, JAX, Ray, GCP (Cloud Training, Data Infrastructure), MLOps, Edge Deployment (ONNX, TVM, TFLite)

EXPERIENCE

Software Architect, Principal Electrical Engineer

Jul 2020 – Present

Aspinity

Pittsburgh, PA

- Transformed novel FPAA chip from academic curiosity into customer-programmable edge AI product with infrastructure unifying compilation, simulation, firmware, and model training/deployment.
- Final authority on embedded software stack including a performance-critical variation-compensating bitstream loader, customer-facing APIs, and real-time test stimulus and measurement DSP system.
- Co-designed development board platform establishing basis for reference designs, customer evaluation, application prototyping, and production semiconductor test.

Principal Electrical Engineer, Solo Founder & CEO

Jan 2016 – Present

Zrna Research

Pittsburgh, PA

- Invented and commercialized platform for “software-defined analog”—reconfigurable embedded hardware that dynamically realizes analog circuits specified in software, in real-time.
- Built third-party compiler toolchain for the Anadigm AN231E04 that maps high-level circuit descriptions onto analog compute fabric (numerical solver, placement, routing, resource allocation).
- Designed and shipped real-time audio DSP hardware platform for musicians and hobbyists.

Software Engineer

Jun 2014 – Jan 2016

Hipmunk (acquired by SAP)

San Francisco, CA

- Pioneered scalable web analytics toolchain that powered ads and marketing teams core workflow.

Contract Software Engineer

Dec 2013 – Jun 2014

Various including Rainforest YC12

San Francisco, CA

- Shipped backend infrastructure for web app acceptance test platform; mobile e-commerce prototype.

Software Engineer

Dec 2012 – Dec 2013

Spotsetter (acquired by Apple)

San Francisco, CA

- Built web crawling and data infrastructure for social location search incorporated into Apple Maps.

Software Engineer (Early Career)

Mar 2009 – Dec 2012

Boston Scientific, FoundationIP

Minneapolis, MN

PATENT

US20250068817A1, *Variation trimming for re-programmable and/or reconfigurable analog circuitry*

EDUCATION

Bachelor of Computer Engineering, Aug 2008 – University of Minnesota, Minneapolis, MN

Undergraduate Honors Thesis: *Computer-Aided Color Appearance Design*